

Nova SBE Econometrics

Class 13
Time Series: Autocorrelation &
Heteroskedasticity

PAULO FAGANDINI

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Nova SBE Econometrics

Testing for Serial Correlation

Link to Slides



Testing for Serial Correlation: Strictly Exogenous Regressors

Order 1 — t -test on lagged residuals

1. Run OLS of y_t on $x_{1t}, \dots, x_{kt}$; obtain residuals $\hat{u}_t$.
2. Regress $\hat{u}_t$ on the one-period lagged residual:

$$\hat{u}_t = \rho \hat{u}_{t-1} + e_t$$

3. Test $H_0 : \rho = 0$ vs. $H_a : \rho \neq 0$ (individual significance).

Order q — joint F -test

1. Run OLS of y_t on $x_{1t}, \dots, x_{kt}$; obtain residuals $\hat{u}_t$.
2. Regress $\hat{u}_t$ on q lagged residuals:

$$\hat{u}_t = \rho_1 \hat{u}_{t-1} + \rho_2 \hat{u}_{t-2} + \dots + \rho_q \hat{u}_{t-q} + e_t$$

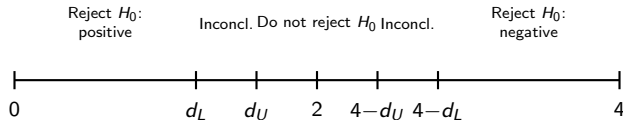
3. Test $H_0 : \rho_1 = \rho_2 = \dots = \rho_q = 0$ vs. $H_a : \text{Not } H_0$.

The Durbin–Watson Statistic

Only detects AR(1) serial correlation. $H_0 : \rho = 0$; $H_a : \rho > 0$.

$$DW = \frac{\sum_{t=2}^T (\hat{u}_t - \hat{u}_{t-1})^2}{\sum_{t=1}^T \hat{u}_t^2} \quad \text{Moreover, } DW \approx 2(1 - \rho)$$

The table provides two critical values d_L (lower) and d_U (upper):



- $DW > d_U \Rightarrow$ fail to reject H_0 (no autocorrelation)
- $DW < d_L \Rightarrow$ reject H_0 (positive autocorrelation)
- $d_L < DW < d_U \Rightarrow$ inconclusive

Without Strictly Exogenous Regressors

Order 1

1. Run OLS of y_t on $x_{1t}, \dots, x_{kt}$; obtain $\hat{u}_t$.
2. Regress $\hat{u}_t$ on lagged residuals *and* explanatory variables:

$$\hat{u}_t = \rho \hat{u}_{t-1} + \delta_1 x_{1t} + \delta_2 x_{2t} + \dots + \delta_k x_{kt} + e_t$$

3. Test $H_0 : \rho = 0$ vs. $H_a : \rho \neq 0$.

Order q

1. Run OLS of y_t on $x_{1t}, \dots, x_{kt}$; obtain $\hat{u}_t$.
2. Regress $\hat{u}_t$ on q lags and all explanatory variables:

$$\hat{u}_t = \rho_1 \hat{u}_{t-1} + \dots + \rho_q \hat{u}_{t-q} + \delta_1 x_{1t} + \dots + \delta_k x_{kt} + e_t$$

3. Test $H_0 : \rho_1 = \dots = \rho_q = 0$ vs. $H_a : \text{Not } H_0$.

Correcting for Serial Correlation (Strictly Exogenous Regressors)

Step 1. Run OLS; get $\hat{u}_t$. **Step 2.** Regress $\hat{u}_t$ on $\hat{u}_{t-1}$ to obtain $\hat{\rho}$.

Cochrane–Orcutt estimator

Apply OLS to $\tilde{y}_t = (1 - \hat{\rho})\beta_0 + \beta_1\tilde{x}_{1t} + \dots + \beta_k\tilde{x}_{kt} + e_t$, where $\tilde{x}_{jt} = x_{jt} - \hat{\rho}x_{j,t-1}$ and $\tilde{y}_t = y_t - \hat{\rho}y_{t-1}$.

Prais–Winsten estimator

Apply OLS to $\tilde{y}_t = (1 - \hat{\rho}^2)^{1/2}\beta_0 + \beta_1\tilde{x}_{1t} + \dots + \beta_k\tilde{x}_{kt} + e_t$, where $\tilde{x}_{jt} = (1 - \hat{\rho}^2)^{1/2}x_{jt}$ and $\tilde{y}_t = (1 - \hat{\rho}^2)^{1/2}y_t$.

Newey–West Standard Errors

Robust to serial correlation (TS5 violated) *and* heteroskedasticity simultaneously.

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Exercises

Exercise 11.5: Wages and Productivity (US, 1960–2005)

Using U.S. data for 1960–2005, the following regression of wages on productivity in the business sector was estimated:

$$\widehat{\ln Comp}_t = 1.6067 + .6522 \ln Prod_t$$

(.0547) (.0124)

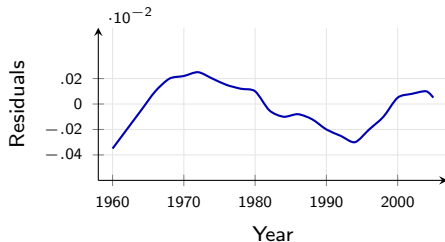
Sub-questions

skip to Exercise 11.6 →

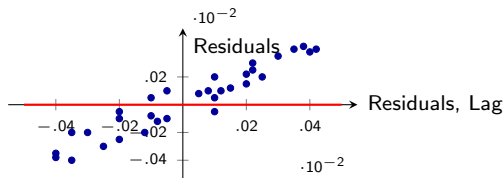
- (a) Interpret residual plots — is there serial correlation?
- (b) Durbin–Watson statistic = 0.218. Conclude?
- (c) Breusch–Godfrey test of order 6. Conclude?
- (d) Trend added to model. Still autocorrelation?
- (e) Three corrections for autocorrelation. Problem solved?
- (f) Heteroskedasticity test. Any solution for both issues?
- (g) Newey–West SEs. Compare with usual SEs.

Exercise 11.5(a): Residual Plots

Residuals over time



Residuals vs. lagged residuals



Conclusion

Plot 1: residuals systematically positive then negative (long runs). Plot 2: residuals and their lags are strongly positively correlated. There is evidence of serial correlation (at least of order 1).

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Exercise 11.5(b): Durbin–Watson Test

$$DW = 0.218, \quad n = 46, \quad k = 1, \quad \alpha = 5\%$$

Hypotheses: $H_0 : \rho = 0$; $H_a : \rho > 0$

From the DW table: $d_L = 1.48$, $d_U = 1.58$.

Decision

Since $DW = 0.218 < d_L = 1.48$, we **reject** H_0 .

There is statistically significant evidence of AR(1) serial correlation.

Why $H_a : \rho > 0$ (one-sided)? The DW statistic is designed specifically to detect positive autocorrelation. Since $DW \approx 2(1 - \rho)$, a very small DW value indicates ρ close to 1, i.e. strong positive correlation.

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Exercise 11.5(c): Breusch–Godfrey Test (Order 6)

Breusch–Godfrey LM test (order 6)

$LM = (n - q)R^2 = (46 - 6) \times 0.7635 = 30.54 > 12.59$ (critical value from χ^2_6 at 5%).
Reject H_0 of no serial correlation. Only $\hat{u}_{t-1}$ is individually significant — suggests AR(1).

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Exercise 11.5(d): Adding a Trend

A time trend was added to the initial model:

$$\widehat{\ln Comp}_t = \underset{(.307)}{.121} - \underset{(.002)}{.008} t + \underset{(.0776)}{1.0283} \ln Prod_t \quad DW = 0.45$$

$n = 46$, $k = 2$, $\alpha = 5\%$: $d_L = 1.43$, $d_U = 1.62$.

Still evidence of AR(1) serial correlation

Since $DW = 0.45 < d_L = 1.43$, we again reject $H_0 : \rho = 0$.

Model misspecification (missing trend) does not explain the autocorrelation.

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Exercise 11.5(e): Correcting for Autocorrelation

Three alternatives were considered:

(i) **Initial model in first differences:** $\Delta \widehat{\ln Comp}_t = .654 \Delta \ln Prod_t, \quad DW = 1.74$
(.057)

(ii) **Model with trend in first differences:** $\Delta \widehat{\ln Comp}_t = .001 + .619 \Delta \ln Prod_t,$
(.003) (.109)
 $DW = 1.71$

(iii) **Cochrane–Orcutt quasi-differenced:** $\ln \widehat{Comp}^*_t = 1.955 + .577 \ln Prod^*_t,$
(.192) (.042)
 $\hat{\rho} = .869, DW = 1.70$

Autocorrelation resolved in all three cases

$n = 45, k = 1, \alpha = 5\%$: $d_L = 1.48, d_U = 1.57$. For all alternatives,
 $(4 - d_U) > DW > d_U \Rightarrow$ fail to reject H_0 . No significant evidence of serial correlation.

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Exercise 11.5(f): Heteroskedasticity Test

White-type test on the initial model:

$$\hat{u}_t^2 = 94.27 - 1.836 \ln(\hat{y}_t) + .009 [\ln(\hat{y}_t)]^2 \quad n = 46, \quad R^2 = 0.296$$

(23.64) (.514) (.003)

LM test for heteroskedasticity

$LM = 46 \times 0.296 = 13.754 > 5.99$ (critical value from χ^2_2 at 5%).

We **reject** H_0 — there is evidence of heteroskedasticity.

Solution for both serial correlation and heteroskedasticity

Newey–West standard errors are robust to both serial correlation and arbitrary heteroskedasticity simultaneously.

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Exercise 11.5(g): Newey–West Standard Errors

Initial model with Newey–West SEs (asterisks denote NW-robust):

$$\widehat{\ln Comp}_t = 1.607 + .652 \ln Prod_t$$

(.079)* (.018)*

Comparison with OLS standard errors

Newey–West SEs are *higher* than the usual OLS SEs (0.079 vs. 0.0547 for the intercept; 0.018 vs. 0.0124 for the slope). This is expected: OLS SEs are too small when serial correlation is present.

The coefficient on $\ln Prod_t$ remains statistically significant:

$$t = \frac{0.652}{0.018} = 36$$

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Exercise 11.6: NYSE Returns and the Efficient Market Hypothesis

A strict form of the **EMH**: information prior to week t should not predict the return during week t . Estimation using weekly NYSE Composite returns (*return*); *L1.return* is the one-period lagged return:

	Coef.	Std. Err.	Obs = 689; F(1,687) = 2.40; $p = 0.1218$		
L1.(return)	.0588984	.0380231	R-squared	=	0.0035
_cons	.179634	.0807419	Adj R-sq	=	0.0020

Sub-questions

skip to Exercise 11.7 →

- (a) Interpret the coefficient. Evidence for the EMH?
- (b) Testing EMH with 2-period lag. How?
- (c) Breusch–Pagan test for heteroskedasticity. Conclude?
- (d) ARCH model for weekly returns?
- (e) ARCH with Gauss–Markov assumptions. Conventional inference?

Exercise 11.6(a): Coefficient Interpretation and EMH

Coefficient interpretation:

$\hat{\beta}_1 = 0.0589$: if the return today increases by one percentage point, the next period return is predicted to increase by 0.06 percentage points, on average, *ceteris paribus*.

Testing the EMH. Under the EMH, past information should not predict current returns. Testing whether weekly returns are correlated with returns from one week prior:

$$H_0 : \beta_1 = 0 \quad H_a : \beta_1 \neq 0$$

Conclusion

$t = 1.55$, $p\text{-value} = 0.122$. We **fail to reject** H_0 even at 10% significance. There is statistical evidence *in favour of* the EMH, even though the estimate suggests a slight positive correlation in weekly NYSE returns.

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Exercise 11.6(b): Testing EMH with Two Lags

To account for possible 2-period correlation, specify:

$$return_t = \beta_0 + \beta_1 return_{t-1} + \beta_2 return_{t-2} + u_t$$

Hypotheses for the EMH:

$$H_0 : \beta_1 = \beta_2 = 0 \quad H_a : \text{Not } H_0$$

Use an F -test of joint significance:

$$F = \frac{R^2/K}{(1 - R^2)/(N - K - 1)} \sim F(K, N - K - 1)$$

Decision rule

We find evidence in favour of the EMH if the p -value is not sufficiently small to reject H_0 , i.e. if past returns are jointly insignificant in predicting current returns.

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Exercise 11.6(c): Breusch–Pagan Heteroskedasticity Test

Squared residuals from the initial model regressed on $return_{t-1}$ ($n = 689$, $R^2 = 0.0419$):

uhatsq	Coef.	Std. Err.	t	$P > t $	[95% CI]	
L1.(return)	-1.104132	.2014029	-5.48	0.000	-1.499572	-.7086932
_cons	4.656501	.4276789	10.89	0.000	3.816786	5.496216

Breusch–Pagan test: $H_0 : \delta_1 = 0$ vs. $H_a : \delta_1 \neq 0$

$t = -5.48$, $p\text{-value} = 0.000$. Reject H_0 at all conventional levels. **Significant evidence of heteroskedasticity.** Volatility in stock returns is *lower* when previous returns were high — variance depends negatively on past returns.

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Exercise 11.6(d): ARCH Model for Stock Returns

Regress $\hat{u}_t^2$ on $\hat{u}_{t-1}^2$ ($n = 688$, $R^2 = 0.1136$, $F(1,686) = 87.92$):

uhatsq	Coef.	Std. Err.	t	P> t	[95% CI]	
L1.	.3370622	.0359468	9.38	0.000	.2664833	.4076411
_cons	2.947434	.4402343	6.70	0.000	2.083065	3.811802

ARCH model: $\hat{u}_t^2 = \gamma_0 + \gamma_1 \hat{u}_{t-1}^2 + e_t$

$H_0 : \gamma_1 = 0$; $H_a : \gamma_1 \neq 0$.

$t = 9.38$, $p\text{-value} = 0.000$. Reject H_0 at all levels. **Strong evidence of ARCH behaviour** in weekly NYSE returns.

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Exercise 11.6(e): Conventional Inference under ARCH

Setup: TS1–TS5 verified; error exhibits **ARCH behaviour**.

Finite-sample inference?

Conventional t/F tests require normality (TS6). ARCH violates normality $\Rightarrow$ finite-sample inference is *not guaranteed*.

Asymptotic inference: TS1'–TS5'

Under TS1'–TS5', t and F statistics are asymptotically valid even without normality. These are the *weaker* conditions sufficient for consistency and asymptotic normality of OLS.

- TS1' Linearity + stationarity & weak dependence
- TS2' No perfect collinearity
- TS3' Contemporaneous exogeneity: $E(u_t | \mathbf{x}_t) = 0$
- TS4' Contemporaneous homoskedasticity: $V(u_t | \mathbf{x}_t) = \sigma^2$
- TS5' No serial correlation: $E(u_t u_s | \mathbf{x}_t, \mathbf{x}_s) = 0, \forall t \neq s$

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Exercise 11.7: Consumption Growth and Income Growth (1973–2009)

Simple regression of real per capita consumption growth (GC) on income growth (GY), 1973–2009:

	Coef.	Std. Err.	Obs= 36; F(1,34)= 71.81; $p = 0.000$		
gy	.570781	.0673545	R-squared	=	0.6787
_cons	.0080791	.0018991	Adj R-sq	=	0.6692

Sub-questions

- (a) Interpret the coefficient of GY . Statistically significant?
- (b) Serial correlation test. Why include GY in the test?
- (c) Evidence for the PIH? Autocorrelation in consumption growth?
- (d) White test for heteroskedasticity. Conclude?
- (e) Joint significance of lagged GY and lagged $i3$?

Exercise 11.7(a): Coefficient Interpretation

$$\hat{\beta}_1 = 0.5708$$

If income growth increases by 1 percentage point, consumption growth increases by 0.571 percentage points, on average, ceteris paribus.

Statistical significance

The t -statistic equals 8.47 and the p -value equals 0.000. Therefore, $\hat{\beta}_1$ is individually statistically significant.

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Exercise 11.7(b): Serial Correlation Test

Test: $\hat{u}_t = \rho \hat{u}_{t-1} + \delta_1 gy_t + e_t$. $\hat{\rho} = -0.119$, $p\text{-value} = 0.566$.

No evidence of AR(1) serial correlation

Fail to reject $H_0 : \rho = 0$ at 5%.

Why include gy in the test?

Strict exogeneity is unlikely to hold; the Breusch–Godfrey approach includes explanatory variables in the auxiliary regression, making the test valid with or without strict exogeneity.

Permanent income hypothesis (PIH): consumption growth is unpredictable, $E(gc_t | \mathbf{I}_{t-1}) = E(gc_t)$ — i.e. $E(gc_t)$ does not depend on the previous value of gc .

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Exercise 11.7(c): PIH Test and Autocorrelation

Regression of gc_t on its own first lag:

gc	Coef.	Std. Err.	t	$P > t $	[95% CI]	
L1.	.446133	.1560472	2.86	0.007	.1286524	.7636135
_cons	.0114307	.0037778	3.03	0.005	.0037448	.0191166

$\hat{\beta}_1 = 0.45$ measures how consumption growth depends on its own previous values.

Testing the PIH: $H_0 : \beta_1 = 0$; $H_a : \beta_1 \neq 0$

$t = 2.86$, p -value = 0.007. We **reject** H_0 at 5% — there is *no evidence in favour of the PIH*. Consumption growth exhibits significant positive autocorrelation (lagged consumption growth predicts current growth).

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Exercise 11.7(d): White Test for Heteroskedasticity

White test: $\hat{u}_t^2 = \delta_0 + \delta_1 gy_t + \delta_2 gy_t^2 + v_t$ ($n = 36$, $R^2 = 0.0979$, $F(2, 33) = 1.79$,

	uhat_sq	Coef.	Std. Err.	t	P> t	[95% CI]	
$p = 0.1828$)	gy	-.000571	.0011288	-0.51	0.616	-.0028676	.0017255
	gy_sq	.0274144	.021115	1.30	0.203	-.0155443	.0703731
	_cons	.0000405	.0000156	2.59	0.014	8.68e-06	.0000723

$H_0 : \delta_1 = \delta_2 = 0$; $H_a : \text{Not } H_0$

$F = 1.79$, $p\text{-value} = 0.18$. We **fail to reject** H_0 at 5%. The model is **homoskedastic**.
No significant evidence of heteroskedasticity.

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Exercise 11.7(e): Joint Significance of Lagged Variables

Model augmented with $L1.gy$ and $L1.i3$ ($i3_t = 3\text{-month T-bill rate}$):

gc	Coef.	Std. Err.	t	$P > t $	[95% CI]	
L1.(gc)	.4443759	.2878617	1.54	0.133	-.1427219	1.031474
L1.(gy)	-.0813176	.1909566	-0.43	0.673	-.4707763	.308141
L1.(i3)	-.0014079	.0007648	-1.84	0.075	-.0029677	.0001519
_cons	.0218872	.0070008	3.13	0.004	.007609	.0361653

$H_0 : \beta_2 = \beta_3 = 0$; $H_a : \text{Not } H_0$; $R_{UR}^2 = 0.2880$, $R_R^2 = 0.1985$.

$$F = \frac{(0.2880 - 0.1985)/2}{(1 - 0.2880)/(35 - 3 - 1)} = \frac{0.04475}{0.02297} = 1.95$$

Conclusion

$p\text{-value} = 0.1596$. Fail to reject H_0 at 5%. $L1.gy$ and $L1.i3$ are *not jointly statistically significant*.

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